

Section A: Introduction & Basics

1. What is the main topic of Lab 05?

- A) Image Classification
- B) Image Segmentation
- C) Object Detection
- D) Image Compression

✓ Correct Answer: B

2. What is the course code for this lab?

- A) AI-4002
- B) AI-4004
- C) CS-4002
- D) CV-4002

✓ Correct Answer: A

3. Which university is this lab manual from?

- A) NUST
- B) FAST
- C) NUCES Karachi
- D) IBA

✓ Correct Answer: C

4. Image segmentation partitions an image into:

- A) Overlapping regions
- B) Distinct, non-overlapping regions
- C) Random pixels
- D) Compressed blocks

✓ Correct Answer: B

5. Each segment in image segmentation corresponds to:

- A) A random color
- B) A meaningful object or part of the scene
- C) A single pixel
- D) The entire image

✓ Correct Answer: B

6. The goal of image segmentation is to separate an image into areas that share similar:

- A) File sizes
- B) Compression ratios
- C) Visual characteristics such as color, texture, or intensity
- D) Metadata

✓ Correct Answer: C

7. Which of the following is NOT a visual characteristic used in segmentation?

- A) Color
- B) Texture
- C) Intensity
- D) File size

✓ Correct Answer: D

8. Preprocessing and feature extraction in segmentation includes:

- A) Pixel values, edge detection, texture analysis, color similarity
- B) File compression, encryption
- C) Audio processing
- D) Text recognition

✓ Correct Answer: A

9. Which of the following is a segmentation algorithm?

- A) Region growing
- B) Clustering
- C) Watershed
- D) All of the above

✓ Correct Answer: D

10. In the landscape example, the segmented region map includes:

- A) Buildings, roads, cars
- B) Sky, mountains, water
- C) People, animals, plants
- D) Text, numbers, symbols

✓ Correct Answer: B

11. In the portrait example, which region is NOT shown?

- A) Hair region
- B) Face region
- C) Clothing region

D) Background region

✓ Correct Answer: D

12. In the street scene example, the segments include:

A) Car, tree, person

B) Sky, water, mountain

C) Hair, face, clothing

D) Cell nuclei, cells

✓ Correct Answer: A

13. Segmentation is the process of dividing an image into:

A) Random parts

B) Meaningful, separate parts

C) Compressed blocks

D) Binary digits

✓ Correct Answer: B

Section B: Mathematics & Real-Life Examples

14. The mathematics behind image segmentation includes:

A) Thresholding, edge detection, clustering, graph theory

B) Calculus, algebra, geometry

C) Sorting, searching, hashing

D) Encryption, decryption

✓ Correct Answer: A

15. Thresholding compares a pixel's intensity value to:

A) A random value

B) A predefined threshold

C) The image mean

D) The maximum pixel value

✓ Correct Answer: B

16. Edge detection algorithms identify:

A) Slow changes in intensity

B) Rapid changes in intensity

C) Constant intensity

D) Random noise

✓ Correct Answer: B

17. Which is a common edge detection method?

A) Sobel

B) Canny

C) Laplacian of Gaussian (LoG)

D) All of the above

✓ Correct Answer: D

18. Clustering techniques group similar pixels together based on:

A) File size

B) Feature similarity

C) Metadata

D) Compression ratio

✓ Correct Answer: B

19. Which of the following is a clustering technique?

A) K-Means

B) Mean-Shift

C) DBSCAN

D) All of the above

✓ Correct Answer: D

20. In graph-based segmentation, pixels are represented as:

A) Edges

B) Nodes

C) Weights

D) Clusters

✓ Correct Answer: B

21. In graph-based segmentation, edges between nodes represent:

A) Distance

B) Similarity between pixel values

C) File size

D) Color depth

✓ Correct Answer: B

22. Which graph algorithms can be used for segmentation?

A) Minimum spanning trees

- B) Graph cuts
- C) Both A and B
- D) Neither A nor B

✓ Correct Answer: C

23. Medical imaging segmentation is used to:

- A) Identify anatomical structures or tumors
- B) Detect pedestrians
- C) Classify land cover
- D) Track objects in AR

✓ Correct Answer: A

24. Autonomous vehicles use segmentation to detect:

- A) Tumors
- B) Cell nuclei
- C) Pedestrians, vehicles, and road signs
- D) Deforestation

✓ Correct Answer: C

25. Satellite imagery segmentation helps to:

- A) Count cells
- B) Classify land cover, monitor deforestation, assess urban growth
- C) Detect tumors
- D) Track objects in real-time video

✓ Correct Answer: B

26. Biomedical imaging segmentation is used to:

- A) Count cells, identify cell nuclei, diagnose diseases
- B) Detect road signs
- C) Classify land cover
- D) Track objects in AR

✓ Correct Answer: A

27. Augmented Reality (AR) uses segmentation for:

- A) Scene understanding and real-time object tracking
- B) Tumor detection
- C) Deforestation monitoring
- D) Cell counting

✓ Correct Answer: A

Section C: Thresholding & Edge Detection

28. Thresholding divides an image into regions based on:

- A) Color depth
- B) Pixel intensity values
- C) File format
- D) Image resolution

✓ Correct Answer: B

29. Global thresholding applies:

- A) Different thresholds to different regions
- B) A single global threshold to the entire image
- C) No threshold
- D) Adaptive thresholds

✓ Correct Answer: B

30. Adaptive thresholding uses:

- A) A single threshold
- B) Different threshold values for different regions
- C) Random thresholds
- D) No threshold

✓ Correct Answer: B

31. Adaptive thresholding is ideal for images with:

- A) Uniform lighting
- B) Varying lighting conditions
- C) No lighting
- D) High contrast

✓ Correct Answer: B

32. Otsu's thresholding automatically selects an optimal threshold to maximize:

- A) Intra-class variance
- B) Inter-class variance
- C) Image contrast
- D) Noise

✓ Correct Answer: B

33. Otsu's thresholding is suitable when the distribution of pixel intensities is:

- A) Uniform
- B) Bimodal
- C) Random
- D) Skewed

✓ Correct Answer: B

34. Color-based thresholding applies thresholding in:

- A) Only grayscale
- B) Multiple color channels (e.g., RGB, HSV)
- C) Only binary
- D) Only alpha channel

✓ Correct Answer: B

35. Color-based thresholding is useful for segmenting objects with:

- A) Similar intensity
- B) Distinct colors
- C) No color
- D) High noise

✓ Correct Answer: B

36. Hysteresis thresholding uses:

- A) One threshold
- B) Two threshold values
- C) Three thresholds
- D) No threshold

✓ Correct Answer: B

37. In Canny edge detection, the high threshold is used to detect:

- A) Weak edges
- B) Strong edges
- C) Noise
- D) Background

✓ Correct Answer: B

38. In Canny edge detection, the low threshold is used to:

- A) Detect strong edges
- B) Link weak edges
- C) Remove noise

D) Increase contrast

✓ Correct Answer: B

39. Hysteresis thresholding is typically used for:

A) Compression

B) Edge detection

C) Color correction

D) Image resizing

✓ Correct Answer: B

40. Edge detection detects boundaries by identifying:

A) Slow changes

B) Rapid changes in intensity or color

C) Constant regions

D) Random noise

✓ Correct Answer: B

41. Edge detection is useful for:

A) Extracting object contours and boundaries

B) Compressing images

C) Enhancing colors

D) Removing noise

✓ Correct Answer: A

Section D: Region Growing & Watershed

42. Region growing starts with a:

A) Random pixel

B) Seed pixel

C) Threshold value

D) Gradient

✓ Correct Answer: B

43. Region growing expands to neighboring pixels that are:

A) Different in intensity

B) Similar in some criteria

C) Random

D) Edges

✓ Correct Answer: B

44. Region growing is effective for segmenting regions with:

A) Complex textures

B) Uniform characteristics

C) High noise

D) Overlapping objects

✓ Correct Answer: B

45. In the region growing code, the seed point is initialized as:

A) (0,0)

B) (10,10)

C) (50,50)

D) (100,100)

✓ Correct Answer: B

46. In the region growing code, the threshold value is:

A) 10

B) 20

C) 50

D) 100

✓ Correct Answer: C

47. In region growing code, the mask is created with dtype:

A) np.float32

B) np.uint8

C) np.int16

D) np.bool

✓ Correct Answer: B

48. Region growing uses a stack for:

A) Sorting

B) Pixel traversal

C) Filtering

D) Compression

✓ Correct Answer: B

49. If the intensity difference is below the threshold, the pixel is:

A) Ignored

B) Added to the segmented region

C) Marked as edge

D) Removed

✓ Correct Answer: B

50. Region growing adds which neighboring pixels?

A) 8-connected

B) 4-connected

C) 16-connected

D) All pixels

✓ Correct Answer: B

51. Watershed segmentation treats the image as a:

A) Binary image

B) Topographic map

C) Graph

D) Histogram

✓ Correct Answer: B

52. Watershed segmentation simulates:

A) Erosion

B) Flooding

C) Dilation

D) Clustering

✓ Correct Answer: B

53. Watershed segmentation is suitable for segmenting:

A) Isolated objects

B) Touching or overlapping objects

C) Binary images only

D) Text

✓ Correct Answer: B

54. In watershed preprocessing, the input image is converted to:

A) RGB

B) Grayscale

C) HSV

D) Binary

✓ Correct Answer: B

55. Marker generation in watershed can be done by:

- A) Manual placement
- B) Thresholding
- C) Distance transform
- D) All of the above

✓ Correct Answer: D

56. Gradient calculation in watershed uses:

- A) Sobel or Scharr filters
- B) Gaussian blur
- C) Median filter
- D) Histogram equalization

✓ Correct Answer: A

57. Marker labeling in watershed uses:

- A) Binary values
- B) Different integer values
- C) Floating-point values
- D) Strings

✓ Correct Answer: B

58. In watershed, high gradient values represent:

- A) Valleys
- B) Peaks/ridges
- C) Flat regions
- D) Noise

✓ Correct Answer: B

59. Boundaries where basins meet represent:

- A) Noise
- B) Segmented regions
- C) Background
- D) Markers

✓ Correct Answer: B

60. Watershed post-processing uses:

- A) Morphological operations
- B) Fourier transform
- C) Histogram equalization

D) Edge detection

✓ Correct Answer: A

61. In watershed code, threshold type used for markers is:

A) THRESH_BINARY

B) THRESH_BINARY_INV + THRESH_OTSU

C) THRESH_TRUNC

D) THRESH_TOZERO

✓ Correct Answer: B

62. Morphological opening removes:

A) Edges

B) Noise

C) Objects

D) Colors

✓ Correct Answer: B

63. In watershed code, sure_bg is created by:

A) Erosion

B) Dilation

C) Opening

D) Closing

✓ Correct Answer: B

64. In watershed code, distanceTransform type is:

A) DIST_L1

B) DIST_L2

C) DIST_C

D) DIST_L12

✓ Correct Answer: B

65. In watershed code, sure_fg threshold is set at:

A) $0.1 * \max$

B) $0.2 * \max$

C) $0.5 * \max$

D) $0.8 * \max$

✓ Correct Answer: B

66. In watershed code, unknown region is calculated as:

A) `add(sure_bg, sure_fg)`

B) `subtract(sure_bg, sure_fg)`

C) `multiply(sure_bg, sure_fg)`

D) `divide(sure_bg, sure_fg)`

✓ Correct Answer: B

67. In watershed code, `connectedComponents` returns:

A) Labels

B) Markers

C) Centers

D) Contours

✓ Correct Answer: B

68. After `connectedComponents`, markers are incremented by:

A) 0

B) 1

C) 2

D) -1

✓ Correct Answer: B

69. In watershed code, unknown region is set to:

A) 1

B) 0

C) -1

D) 255

✓ Correct Answer: B

70. In watershed, boundaries are marked as:

A) 0

B) 1

C) -1

D) 255

✓ Correct Answer: C

71. In watershed code, boundaries are colored:

A) `[0,0,255]`

B) `[255,0,0]` red

C) `[0,255,0]` green

D) `[255,255,255]` white

✓ Correct Answer: B

Section E: Clustering, Deep Learning & Active Contours

72. Clustering-based segmentation groups pixels based on:

- A) File size
- B) Feature similarity
- C) Metadata
- D) Compression

✓ Correct Answer: B

73. Which is a clustering technique?

- A) K-Means
- B) Mean-Shift
- C) DBSCAN
- D) All of the above

✓ Correct Answer: D

74. Feature extraction for color images can use:

- A) RGB
- B) LAB
- C) HSV
- D) All of the above

✓ Correct Answer: D

75. Cluster initialization decides:

- A) Number of clusters K
- B) Number of pixels
- C) Image size
- D) Threshold

✓ Correct Answer: A

76. In K-Means, the assignment step assigns each pixel to:

- A) Random cluster
- B) Nearest cluster centroid
- C) Farthest cluster
- D) All clusters

✓ Correct Answer: B

77. In K-Means, the update step recalculates centroids as:

- A) Median of pixels
- B) Mean of pixels in each cluster
- C) Mode of pixels
- D) Random values

✓ Correct Answer: B

78. K-Means convergence is reached when:

- A) Centroids are random
- B) Cluster assignments no longer change significantly
- C) All pixels are in one cluster
- D) Image is compressed

✓ Correct Answer: B

79. After convergence, each cluster represents:

- A) Noise
- B) A segment or region
- C) An edge
- D) A pixel

✓ Correct Answer: B

80. Post-processing in clustering can involve:

- A) Merging or splitting clusters
- B) Compression
- C) Encryption
- D) Resizing

✓ Correct Answer: A

81. In K-Means code, pixel_values is reshaped to:

- A) (-1, 2)
- B) (-1, 3)
- C) (-1, 4)
- D) (-1, 1)

✓ Correct Answer: B

82. In K-Means code, pixel_values is converted to:

- A) np.uint8
- B) np.float32
- C) np.int16

D) np.bool

✓ Correct Answer: B

83. In K-Means code, criteria is:

A) TERM_CRITERIA_EPS + TERM_CRITERIA_MAX_ITER, 100, 0.2

B) TERM_CRITERIA_EPS, 50, 0.1

C) TERM_CRITERIA_MAX_ITER, 200, 0.5

D) None

✓ Correct Answer: A

84. In K-Means code, K value is:

A) 2

B) 3

C) 4

D) 5

✓ Correct Answer: C

85. In K-Means code, flags used is:

A) KMEANS_PP_CENTERS

B) KMEANS_RANDOM_CENTERS

C) KMEANS_USE_INITIAL_LABELS

D) None

✓ Correct Answer: B

86. In K-Means code, centers are converted to:

A) np.float32

B) np.uint8

C) np.int16

D) np.bool

✓ Correct Answer: B

87. Deep learning-based segmentation uses:

A) CNNs

B) RNNs

C) GANs

D) SVMs

✓ Correct Answer: A

88. Semantic segmentation means:

A) Different instances of same class separated

B) Same class gets same label

C) Random labels

D) No labels

✓ Correct Answer: B

89. Instance segmentation means:

A) Same class gets same label

B) Different instances of same class are separated

C) No labels

D) Binary labels

✓ Correct Answer: B

90. Deep learning segmentation is suitable for:

A) Simple tasks

B) Complex tasks and large datasets

C) Small datasets only

D) Binary images only

✓ Correct Answer: B

91. Active contour models are also called:

A) Snakes

B) Birds

C) Worms

D) Fish

✓ Correct Answer: A

92. Active contour models evolve based on:

A) Random motion

B) Image gradients and constraints

C) Histogram

D) Fourier transform

✓ Correct Answer: B

93. Active contour models are effective for:

A) Object tracking and boundary refinement

B) Compression

C) Color correction

D) Noise removal

✓ Correct Answer: A

Section F: OpenCV Functions

94. cv2.imread() flag for grayscale is:

- A) IMREAD_COLOR
- B) IMREAD_GRAYSCALE
- C) IMREAD_UNCHANGED
- D) IMREAD_ANYCOLOR

✓ Correct Answer: B

95. cv2.threshold() returns:

- A) Only binary mask
- B) retval and dst
- C) Only retval
- D) None

✓ Correct Answer: B

96. Otsu thresholding is combined with:

- A) THRESH_TRUNC
- B) THRESH_BINARY + THRESH_OTSU
- C) THRESH_TOZERO
- D) THRESH_BINARY_INV only

✓ Correct Answer: B

97. cv2.cvtColor() code for BGR to HSV is:

- A) COLOR_BGR2GRAY
- B) COLOR_BGR2HSV
- C) COLOR_BGR2RGB
- D) COLOR_BGR2LAB

✓ Correct Answer: B

98. cv2.inRange() parameters are:

- A) src, lowerb, upperb
- B) src, threshold
- C) src, kernel
- D) src, markers

✓ Correct Answer: A

99. cv2.Canny() parameters are:

- A) image, threshold1, threshold2
- B) image, K
- C) image, markers
- D) image, kernel

✓ Correct Answer: A

100. cv2.kmeans() returns:

- A) compactness, labels, centers
- B) retval, dst
- C) markers
- D) edges

✓ Correct Answer: A

101. cv2.watershed() marks boundaries as:

- A) 0
- B) 1
- C) -1
- D) 255

✓ Correct Answer: C

102. cv2.adaptiveThreshold() parameter for neighborhood size is:

- A) C
- B) blockSize
- C) maxValue
- D) thresholdType

✓ Correct Answer: B

103. cv2.morphologyEx() with MORPH_OPEN performs:

- A) Dilation then erosion
- B) Erosion then dilation
- C) Only erosion
- D) Only dilation

✓ Correct Answer: B